

employed in each ISCO-08 1-digit group and from expert opinions. As the final step, we provide updated global, regional and income-based estimates of employment that has the potential to be transformed by Gen AI technology.

Our study specifically targets GenAI tools, isolating the effects of this recent family of AI from broader AI applications, such as the use of Machine Learning (ML) algorithms or image recognition in industrial production. There are several considerations that justify such selective focus. First, we respond to the growing need for a research tool that could provide a better picture of GenAI's potential impact on national labour markets, given the growing societal angst concerning this technology and the recent reports about increasing levels of adoption of GenAI at work (Bick et al., 2024; Faverio and Tyson, 2022; Grampp et al., 2023; Maison & Partners and ThinkTank, 2024; Pew Research Center, 2023; Rutgers, 2024). Second, while several indicators of occupational exposure to broader AI technologies are available to researchers (Nurski and Vansteenkiste, 2024), very few tools allow for a more detailed focus on GenAI (Nurski and Ruer, 2024). A new, more precise exposure index enables a closer alignment of the academic work on digital economy and labour with the significant interest in the GenAI technology in the media and in the public debate.¹ Third, the methodological blueprint provided by our research can be quite flexibly expanded to other types of AI in the future or focused more narrowly on specific subsets of digital technologies and sectors of their application.

This paper is the first in a series of upcoming publications based on ILO and NASK cooperation that strive to improve the methodology for assessing employment effects of AI, but also to document in detail the methods used, including the survey, expert verification process and the construction of the AI model used for predictions. Subsequent papers will present an assessment of the potential impact of GenAI on the Polish labour market, based on more detailed occupation and task-level data and using a mix of quantitative and qualitative methods. All publications will include the methodological blueprints and technical details of both the quantitative and qualitative components, including survey questionnaires and interview guidelines. By making these tools openly accessible, we hope to contribute to the development of an improved method for assessing potential impact of GenAI on jobs and to stimulate more detailed national-level studies, including in lower- and middle-income countries.

¹ For example, Felten et al. (2023) provided an updated version of their original exposure index from 2021, adjusting for the abilities of GenAI tools. As shown in their paper, the adjustment does not result in a major shift in occupations exposure and still makes it impossible to assess the abilities of this new technology in a selective manner. See Gmyrek et al. (2024) for a more detailed comparison of these scores.

► 1 Task-based approaches to automation in the literature

Central to the study of technology on work is the insight that jobs are a “bundle of tasks” (Autor, 2015). As such, task automation might, or might not lead to job automation, depending on the importance of a particular task to an occupation. Analyses of employment effects of automation that use the task-based approach, including this study, attempt to capture two central considerations: (i) the ability of a specific technology to automate a given task and (ii) the character of existing occupations in the labour market, based on the detailed tasks performed by these occupations. By modelling the intersection of these two dimensions, one arrives at a range of possible projections of how such technology could interact with occupational tasks, leading either to their full automation through technology-driven substitution, or to a transformation, through partial automation and human-machine complementarity. While the task-based framework certainly has limitations, recent academic literature shows the analytical advantages of this method for modelling employment and labour market impacts (Acemoglu et al., 2024; Nurski and Ruer, 2024).

Research on AI and tasks has shown important improvements in methods and scope in recent years, which includes using AI tools for research purposes. In one of the early papers on Machine Learning (ML) and jobs, Frey and Osborne, (2013) fed the opinions of a group of IT technology experts into an ML algorithm to develop occupational-level projections of automation potential for the US job market. The emergence of GenAI APIs has simplified the possibility of combining human expertise and algorithmic capacities for research purposes. Relying on this idea, Eloundou et al. (2023) demonstrated a close alignment of GPT-4 predictions with a survey of 70 AI experts on the potential of automating occupational tasks with LLMs, and subsequently built on this feature to develop synthetic automation scores for the US market.

Most task-based studies are focussed on the US, due to the very detailed nature of the O*NET database, as well as its public accessibility. This typically forces researchers interested in the global picture or other countries and regions to base their analysis on the strong assumption of similarity of local economic and labour effects to those projected in the US context. In reality, occupational tasks can vary significantly among individual countries, which poses a particular challenge in developing countries. As such, working with tasks in the ISCO-08 system offers a useful and simple common denominator for projections across different countries, regions and income groups. Our method combines the strengths of this global framework with a significantly more detailed national classification system in Poland, which provides a detailed list of occupations and tasks, based on the latest labour market data. This helps us establish a starting point for modelling interaction of GenAI’s abilities with up-to-date tasks and occupations in a setting located between the countries with highest incomes and the emerging economies (Figure 1). Using a combination of GenAI tools, we subsequently transfer this knowledge back to the tasks in ISCO-08 system.

► Conclusion

In this paper, we have presented the methodology used to develop a new global index of occupational exposure to GenAI, as well as new estimates of the global effects on employment. We build on the approach developed by the ILO in 2023, but with several notable refinements. First, by beginning our analysis with the Polish 6-digit occupational structure, we expand by ten-fold the amount of detailed task information available for the analysis of task automation. In addition, using a representative sample from the 29,753 tasks in the Polish occupational classification system and a survey of 1,640 people in Poland, we captured the opinions of each 1-digit ISCO-08 group on the potential for task automation in the job categories in which they are currently employed (52,558 data points). We then compared this input with an expert survey and conducted subsequent rounds of revisions among a smaller international expert group for the tasks with the highest degree of disagreement. Based on this process, we built a repository of knowledge with adjusted task scores that goes beyond Poland's national specificities and reflects a broader consensus on the task automation potential of current GenAI tools.

We subsequently used this repository to develop an AI assistant that could predict scores for these tasks based on this additional information as well as the technical documentation of ISCO-08. These scores update the original set of scores produced by the ILO in 2023 (Gmyrek et al., 2023). The use of a verified “knowledge base” presents an important methodological improvement, which greatly enhances stability of our predictions across independent LLMs (see Section 6), and which can be used to produce similar score predictions for other occupational classification systems that contain job-specific tasks. In addition, we make all task-level scores publicly available, which also makes it possible for researchers to organize the occupational exposure in different ways than our proposed framework of exposure gradients.

The new global index presented in this paper is generic, due to the nature of ISCO-08 and the structure of tasks associated with occupations in this framework. Its definite strength – similar to ISCO-08 – is that it provides a common global denominator and a direct link to national labour forces surveys held in the ILO harmonized micro data repository, which relies on the same classification of occupational structures. Even though all countries do not use ISCO-08, mappings of the national systems to this common international standard are available. This eliminates the need for crosswalks from other national systems (e.g. O*NET), which involves tenuous assumptions about the similarity of detailed occupations in the technologically advanced context of the USA to other countries, especially non-HICs. The simple and comparable character of our ISCO-08 based index enables direct production of global, regional and income-based employment estimates, based on the same statistical modelling techniques that are used by the ILO for other global figures (see Section 11).

Our 2025 scores are presented in a revised occupational classification system, adopting a framework of four progressively increasing exposure gradients, that better highlights the importance of potential AI exposure and task variability for determining possible outcomes. We produced a new set of global estimates of GenAI exposure based on this revised classification.

The new index maintains many of the headline findings of the 2023 methodology. Clerical occupations continue to have the highest exposure levels, albeit with some important downward revisions to individual tasks scores. Such occupations include data entry clerks, typists, accounting, bookkeeping clerks and administrative secretaries. Additionally, some occupations – such as financial analysts, and web and multimedia developers (gradient 4), as well as bank tellers, translators, applications programmers, investment advisers (gradient 3) – have increased their